

$$f(x) = \begin{cases} 1 & \text{dla } x \in (0, a) \\ 0 & \text{dla } x \in (a, \pi) \end{cases}$$

$$a_0 = \frac{2}{\pi} \left( \int_0^a 1 dx + \int_a^{\pi} 0 dx \right)$$

$$[x] \Big|_0^a = a$$

$$a_0 = \frac{2}{\pi} (\cdot a) = \frac{2a}{\pi}$$

$$a_n = \frac{2}{\pi} \left( \int_0^a \cos \frac{n\pi x}{\pi} dx + \int_a^{\pi} 0 \cdot \cos 2nx dx \right)$$

$$\int_0^a \cos 2nx = \left[ \frac{1}{2n} \sin 2nx \right] \Big|_0^a = \frac{1}{2n} \sin(2an) - 0$$

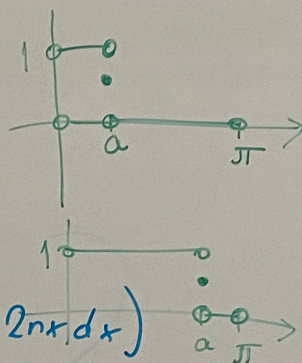
$$a_n = \frac{2}{\pi} \left( \frac{1}{2n} \sin 2an \right)$$

$$a_n = \frac{2}{2\pi n} \sin 2an$$

$$T = \pi$$

$$\frac{T}{2} = \frac{\pi}{2}$$

$$\frac{1}{\frac{\pi}{2}} = \frac{2}{\pi} = \frac{1}{\frac{\pi}{2}}$$





$$b_n = \frac{2}{\pi} \left( \int_0^a \sin \frac{n\pi x}{2} dx + \int_a^{\pi} 0 \cdot \sin \frac{n\pi x}{2} dx \right)$$

$$b_n = \frac{2}{\pi} \left( \int_0^a \sin 2nx dx \right) \quad \int \sin 2nx = \left[ -\frac{1}{2n} \cos 2nx \right]_0^a$$

$$b_n = \frac{2}{\pi} \left( \frac{1}{2n} (1 - \cos 2an) \right) \quad \leftarrow (1)^n = -\frac{1}{2n} \cos 2an + \frac{1}{2n} (\cos 0)$$

$$b_n = \frac{2}{2\pi n} (1 - \cos 2an)$$

$$f(x) \sim$$

$$\frac{\frac{2a}{\pi}}{2} + \sum_{n=1}^{\infty} \frac{1}{\pi n} \sin 2an \cdot \cos \frac{n\pi x}{2} + \frac{1}{\pi n} (1 - \cos 2an) \sin \frac{n\pi x}{2}$$

$$\frac{a}{\pi} + \frac{1}{\pi n} \sum_{n=1}^{\infty} \sin 2an \cos 2nx + (1 - \cos 2an) \sin 2nx$$